Magnetic Matter Movers Phase 0 — Execution Checklist & Gate Criteria

Buy → Build → Measure → Prove → Unlock Phase 1

Companion to Build Sheets 001 (Inductrack) and 002 (Droplet Thruster). Print this; fill it in ink. The filled document + photos/video is your gate evidence — bring it back and we review it together, line by line.

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0. How the Gate Works

Phase 0 closes when **both** builds pass their gates below. “Pass” means: the actual-vs-predicted tables are filled, deviations are explained (not ignored), and the evidence artifacts exist. Deviating from prediction does **not** fail the gate — *unexplained* deviation does. A 2× miss with a good story is data; a perfect match with no photos is nothing.

Gate evidence package (what you bring back): ☐ Filled tables from §3 and §5 (photo of this document is fine) ☐ Video: 001 hovering ≥60 s · 002 displacement + snap-back ☐ Photos: Halbach jig mid-assembly · enclosure interior with beam dump · goggles on head ☐ Anomaly notes: one paragraph per deviation >50% from prediction

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1. Purchase Checklist (order everything at once — one shipping wait, not four)

Build 001 — Inductrack (~$80–120) ☐ 10× ½” N42 cube magnets (8 + 2 spares) ☐ Alloy-rim bike wheel w/ axle (Variant A) — used is fine, must be aluminum rim (magnet test: doesn’t stick) ☐ 775-class motor or brushless + ESC, ≥50 W, with speed control ☐ O-ring/belt for friction drive · epoxy (JB Weld) · 2× guide rods + PTFE tape ☐ Print queue: magnet jig, cart body (STL/design per Sheet 001 §5)

Build 002 — Droplet (~$150) ☐ TinyLev kit or parts (72× 40 kHz transducers, Arduino Nano, L298N, frame print) ☐ ≤5 mW red laser (starter) — this is the ONLY laser you need to pass Gate 002 ☐ 100–500 mW 450 nm module + **OD4+ goggles rated 445–455 nm** (may defer both to post-gate) ☐ 1 mL syringe + blunt 25 ga needle · india ink · foamcore/felt for enclosure + beam dump ☐ Optional: photodiode + resistor ($2) · laser power meter ($25)

Instruments (shared) ☐ Feeler gauges · phone tripod · mm-grid card (print one) · multimeter with current mode (or clamp meter) ☐ Phone apps: tachometer (acoustic or strobe), slow-mo camera

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2. Build 001 Sequence (checkbox version of Sheet 001 §5–6)

☐ Jig printed; ONE magnet test-fitted dry ☐ All 10 magnets polarity-marked (red dot = N face) BEFORE any assembly ☐ Array assembled in jig, one cube at a time, pusher block not fingers, clamped through full epoxy cure ☐ **Strong-side check: washer snaps hard to one face, barely holds on the other — strong face marked** ☐ Wheel mounted, spins smoothly to 1,500 RPM, plywood guard over lower half ☐ Guide rails mounted: cart free vertically ~25 mm, constrained laterally ☐ Array epoxied strong-face-down into cart; ballast to ~250 g total (record actual: ______ g)

3. Build 001 — Actual vs. Predicted

Table A — Levitation onset & gap (Variant A predictions; B in parentheses)

Measurement	Predicted	Actual	Dev.
First cart motion (RPM)	~40–60 (40)	______	___
Clean liftoff (RPM)	~60 (54)	______	___
Gap @ 500 RPM (mm)	1–3 (3–6)	______	___
Gap @ 1,000 RPM (mm)	2–6 (6–12)	______	___
Gap @ 1,500 RPM (mm)	3–8 (8–15)	______	___

Table B — The flat-drag signature (motor input watts = V × A, minus no-cart baseline at same RPM)

RPM	Baseline W (no cart)	W with cart	Drag W (diff)
500	______	______	______
1,000	______	______	______
1,500	______	______	______

Predicted: drag column ≈ flat, 30–60 W (A) / ~28 W (B). **The plateau in Table A and the flat line in Table B are the physics**

deliverables.

Troubleshooting triage (if outside 2× of prediction): - No lift ever → strong-side check failed or array glued weak-side-down → rebuild with spares - Liftoff RPM ≫ predicted → rim wall thinner than assumed (steel-belted or thin rim) or gap-robbing cart tilt → verify rim is aluminum, check rail squareness - Gap far below prediction → cart overweight (weigh it) or rim conductivity low → reduce ballast, log both - Drag not flat → you're reading total motor W not the difference → redo baseline column

GATE 001 — pass conditions: ☐ Stable hover ≥60 s at ≤1,500 RPM (video) ☐ Table A filled, gap plateaus with RPM ☐ Table B filled, drag flat within ±30% over a 2–3× speed range ☐ Any >50% deviation has an anomaly paragraph

4. Build 002 Sequence (checkbox version of Sheet 002 §5–6)

☐ TinyLev assembled; **foam bead traps rock-solid** (do not proceed to liquid until true) ☐ Enclosure built: grid card behind trap, camera port, horizontal beam path into felt dump ☐ ≤5 mW red laser rigidly mounted, aligned through node using trapped bead, locked down ☐ Dyed water mixed (~1 drop ink/mL); droplet loading practiced until 3 consecutive stable loads ☐ IF proceeding to 3B later: enclosure interlock, goggles for everyone present, key control — photo evidence required before any high-power run

5. Build 002 — Actual vs. Predicted (all at ≤5 mW; camera-tracked)

Table C — Trap stiffness calibration (tilt method, Sheet 002 §7.2)

Tilt angle	Force mg·sinθ (μN)	Displacement (mm)	k (μN/mm)
2°	1.4	______	______
4°	2.9	______	______
6°	4.3	______	______

Predicted k: 1–10 μN/mm. Three rows should agree within ~30% (that's your instrument error bar).

Table D — Thrust runs (displacement × k = thrust; laser incident 5 mW, absorbed unknown → that's what you're measuring)

Run	Displacement (mm)	Thrust (μN)	Direction horizontal?	Snap-back on off?
1	______	______	Y / N	Y / N
2	______	______	Y / N	Y / N
3	______	______	Y / N	Y / N

Predicted thrust at 5 mW incident: 0.15–0.4 μN (absorption 30–80%). Implied absorption fraction = measured thrust ÷ 0.5 μN: ______

Troubleshooting triage: - No displacement → underdyed (add ink), beam missing droplet (re-align on bead), or displacement below camera resolution (zoom, track centroid frame-by-frame) - Vertical displacement → convection or beam above/below node → re-align; beam must pass through node center - Droplet dies instantly → too big or trap overdriven → smaller droplet, lower amplitude - k values disagree wildly → trap drifting with temperature → let it warm up 5 min, recalibrate

GATE 002 — pass conditions: ☐ Table C filled, k self-consistent within ~30% ☐ Table D: 3 runs, horizontal direction confirmed, snap-back confirmed (video of one run) ☐ Implied absorption fraction between 0.1 and 1.0 (sanity bound) ☐ Any >50% deviation has an anomaly paragraph ☐ *Stretch (not required):* offset map (Sheet 002 §7.6) — 5 offsets, force-vs-offset sketch

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6. What Passing Unlocks — Phase 1

Bring the evidence package back to this chat. We review deviations together (that conversation IS the troubleshooting session), then Phase 1 opens with, in order:

1. **Lab Note 001** — we co-write it from your filled tables: the thrust-vs-power result, the flat-drag result, methods, photos. First publishable Magnetic Matter Movers artifact.
2. **High-power droplet campaign** — 3B laser unlocked (safety photo evidence first): full thrust-vs-power curve at 4 power levels, limit-cycle hunt, offset map if not done.
3. **Wave-optics propagation sim** — the winter project; the go/no-go computation on beam delivery. Built together, here.
4. **Build Sheet 003 scoping** — torsion-pendulum thrust stand, the bridge toward the combined setup.

Rule of the ladder: no step skips. The gate exists because Phase 1's credibility rests on Phase 0's error bars.